

Krishnaa Vadivel | Software Engineer | Machine Learning | Distributed Computing

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Profile

Software engineer and PhD researcher focused on machine learning, distributed computing, numerical software, and engineering tooling. Strong background in Python and C++ development for simulation, automation, GPU acceleration, data systems, and infrastructure spanning research and production-adjacent environments.

Research Experience

Yale University, Electrical Engineering

Research Assistant / PhD Researcher

2021–present

- Develop scientific software for first-principles and many-body workflows in materials research, combining Python and C++ with MPI, OpenMP, PETSc, SLEPc, CUDA, ROCm, and Linux-based automation.
- Build distributed and accelerator-enabled workflows for HPC environments including Frontier, Frontera, and Perlmutter, with experience spanning NVIDIA and AMD GPU systems on national-lab clusters.
- Create reusable tooling, notes, and structured workflows that bridge theory, implementation, debugging, and reproducibility in computational science.
- Extend into ML-assisted and retrieval-driven automation where useful, particularly with LangChain, OpenAI API models, and local Ollama models for coding and research-workflow setup.
- Co-authored a 2025 *Journal of the American Chemical Society* paper and presented APS talks in 2024, 2025, and 2026 on self-trapped exciton formation and related first-principles work.

Professional Experience

Probe Technologies Inc. / Weatherford Wireline Services

Software and Embedded Systems Engineer

2019–2021

- Built CUDA-accelerated processing software for large acoustic waveform and sensor datasets used in engineering diagnostics and field-tool analysis.
- Developed ASP.NET services and Microsoft SQL Server workflows for internal software, field-data management, and remote tool-calibration support.
- Developed C# GUI applications for preparing customer inputs and interacting with engineering workflows tied to deployed hardware systems.
- Worked close to embedded and instrumentation systems, giving practical experience with software operating alongside deployed hardware and field constraints.

FindLight

Photonics Technical Writer

2018–2019

- Produced technical content for optics and photonics topics, demonstrating strong technical communication across specialized engineering subject matter.

Selected Projects

Equivariant Graph Neural Networks for Molecular Modeling

- Built PyTorch Geometric workflows for molecular-property prediction using equivariant graph neural networks and graph-transformer-style modeling ideas on QM9-style data.

Agentic Coding and Scientific Workflow Generation

- Developed LangChain-based retrieval-driven tooling using OpenAI API models and local Ollama models to read documentation and academic papers, then generate structured starter inputs and workflow artifacts.

Scientific Visualization and GPU Tooling

- Built CUDA-based tooling and related utilities for large numerical datasets, interactive inspection, and engineering communication.

Photonics and Simulation Code Artifacts

- Created clean, inspectable software artifacts for FDTD-based photonics simulation and related numerical-science workflows.

Education

Yale University

PhD, Electrical Engineering

2021–present

Ongoing work combining scientific computing, high-performance workflows, and computational materials research.

Yale University

MPhil, Electrical Engineering

2023

Yale University

MS, Electrical Engineering

2023

Cornell University

MEng, Electrical and Computer Engineering

2017–2018

Cornell University

BS, Electrical and Computer Engineering

2013–2017

Selected Coursework

Computing Foundations: Theoretical Computer Science; Probabilistic Machine Learning

Scientific Foundations: Graduate Quantum Mechanics; Solid State Physics; Many-Body Quantum Physics

Engineering Foundations: Signals and systems, embedded systems, semiconductor physics, and numerical/scientific computing practice

Technical Skills

Languages: Python, C, C++, JavaScript, Bash, PowerShell, SQL

Scientific Computing: MPI, OpenMP, PETSc, SLEPc, mpi4py, petsc4py, slepc4py, NumPy, pandas, SciPy, SymPy, matplotlib

Systems: Linux command line, CMake, Docker, Docker Compose, Kubernetes, gcloud

Platforms: CUDA, ROCm, Frontier, Frontera, Perlmutter

ML / Automation: PyTorch, PyTorch Geometric, scikit-learn, LangChain, OpenAI API, Ollama, graph neural networks, graph transformers, agentic workflows